



PROFILE

Year 13 student (Maths, Computer Science, Product Design) and part-time Junior Software Engineer with a real love of combining hardware and software, in my coursework and personal projects alike. I run a Docker-based home lab with a Zigbee smart home and a Jellyfin media server, built an automated pet feeder for my EPQ, and I'm an RAF Air Cadets Sergeant. Seeking a software or engineering degree or higher apprenticeship starting September 2027.

KEY SKILLS

Leadership	Air Cadets Sergeant responsible for junior cadets; Scouts Young Leader
Communication	Plan and deliver lessons to groups of 10–20; Instructor First Aid
Teamwork	Team project on work experience at MBDA; part-time at Janus Technology; volunteer teams at events
Languages	C#, Python, JavaScript
Web & software	React, SvelteKit, HTML/CSS; building web apps and plugins to a brief
Infrastructure	Docker, Linux, self-hosting, home networking, automated image builds and deploys
AI tooling	LLM-based automation, Model Context Protocol (MCP) servers, structured knowledge bases
Hardware	Raspberry Pi, Zigbee, LoRa, sensors and smart devices, wiring and low-voltage installation, hardware integration, testing and fault-finding
Design	Product design from brief to prototype: CAD, 3D printing and electronics (A-Level Product Design, GCSE DT grade 9)

EDUCATION

Knights Templar School Sixth Form Baldock, Hertfordshire	Sept 2024 – present
A-Levels (predicted): Product Design A*, Computer Science A, Maths B Extended Project Qualification — A* (50/52): designed and built an automatic pet feeder, covering the research, design, electronics and build. Full write-up on my portfolio. GCSEs: grade 9 in Design and Technology and Computer Science; 8 in Physics; 7 in Maths; 5 in English; plus 5 further GCSEs at grades 6–8 and a Cambridge National in Enterprise and Marketing (M2, grade A equivalent). Graded coursework is on my portfolio.	

WORK EXPERIENCE

Janus Technology Junior Software Engineer (part-time)	July 2025 – present
Support ticket automation (osTicket) — built a plugin for the open-source osTicket helpdesk that reads each incoming ticket, works out the likely problem and drafts a step-by-step reply, so common queries get a useful first response without waiting for an agent. Stile app for AVProEdge — added the live video preview feature to a released app that controls multiview video walls. Every draggable source now shows a real-time preview, so users can tell inputs apart before assigning them to a display. AI knowledge base — set up a linked Obsidian vault holding API references, example projects and in-house library notes so AI tools can pull the right context instead of the team pasting docs into a chat window. Also tested several MCP servers and wrote a custom one to connect AI tools to our self-hosted GitLab. - Deliver each project to a brief, mostly independently within a small team, revising it after code review and feedback from colleagues.	
MBDA Engineering Work Experience	2026
Miniature guided missile — in a team, designed and built a model guided missile running on a Raspberry Pi. I wrote part of the control code, wired in the hardware, tested the system and tracked down faults. Code on GitHub at JamesDillonDev/mini-missile. - Rotated through systems testing and validation, materials, hardware-in-the-loop, electronics, production and project management, which showed me how a large engineering programme is delivered from concept through to implementation.	

LEADERSHIP & VOLUNTEERING

RAF Air Cadets, 248 Squadron | Sergeant

2022 – present

- Promoted through four ranks to Sergeant; train and look after junior cadets every week on parade nights.
- Methods of Instruction qualified: I plan and deliver lessons to groups of 10–20 and assess cadets afterwards. Hold the Silver Leadership award.
- Qualified First Aid Instructor (St John Ambulance), able to teach and assess first aid as well as respond to incidents.
- Represent the squadron at public events such as Remembrance Sunday, and competed in athletics at Wing and Regional level.

Scouts and Explorers | Young Leader

2014 – present

- Plan and run activities at weekly meetings of my old Scout group, keeping sessions on track and supporting the leaders.
- Organise kit, give safety briefings and run outdoor activities on weekend camps and residentials.
- Volunteer on the BBQ and bar at Balstock and the Baldock Beer Festival, serving 100+ members of the public, handling cash and keeping a busy food station clean and safe.

VEX Robotics Club | Team Member & Mentor, Knights Templar School

Sept 2026 – present

- Design and build robots for VEX competition tasks such as moving and stacking objects, competing against other school teams in head-to-head arena matches.
- In the main team I help design mechanisms and parts. I also mentor younger groups, teaching the basic ideas and concepts of robot design.

Duke of Edinburgh's Award | Bronze completed, Silver in progress

2022 – present

- Expeditions in North Yorkshire and the Peak District involving route planning, navigation and teamwork, volunteering in the local community, and a cybersecurity club covering online safety and ethical hacking.

PROJECTS

OpenHighways — Live UK Traffic Camera Map | React, Leaflet, Docker

2026

- **openhighways.uk** — a free, live map of UK traffic cameras, pulling feeds from National Highways, TfL, Traffic Scotland, Traffic Wales and TrafficWatchNI into one place, with on-device vehicle counting for busy roads. Code on GitHub at [JamesDillonDev/openhighways](https://github.com/JamesDillonDev/openhighways).
- Self-hosted and deployed the same way as my portfolio site, with a Docker image rebuilt and redeployed automatically on every push.

jamesdillon.uk — Portfolio Website | SvelteKit, Docker, Raspberry Pi

2025 – present

- Built the site in SvelteKit and self-host it from a custom Docker image. Pushing to Git builds a new image that my server pulls and swaps in automatically, so a deploy takes nothing more than a git push.
- Wrote a custom in-page PDF viewer component so my coursework can be read on the site instead of having to be downloaded first.

Home Lab — Home Assistant Smart Home & Jellyfin | Docker, Zigbee, Raspberry Pi

2025 – present

- Rebuilt the family smart home around Home Assistant in Docker. Started with Tapo smart plugs for lighting, then added Wake-on-LAN control for the TV and Sonos playback.
- Moved over to Zigbee using a Sonoff dongle and motion sensors for automations like hallway lights that come on at night and switch themselves off, including planning cable routes and running low-voltage cabling safely and neatly.
- Added a wall-mounted Raspberry Pi 4 touchscreen panel so the whole house can use the system without reaching for a phone. I diagnose and fix the hardware, network and integration problems myself.
- Also run a Jellyfin media server so the family has one central library, including the CDs I have ripped, simple enough for everyone to use without my help.

QUALIFICATIONS & INTERESTS

Qualifications Instructor First Aid (St John Ambulance) · Methods of Instruction · BTEC Level 2 Teamwork and Personal Development · Fire Warden

Interests Running (5K PB 20:35, Wing-level athletics gold), home lab and self-hosting, 3D printing and electronics projects

Full write-ups, coursework and source code: jamesdillon.uk